

Hayato Isa

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EDUCATION

University of California, Berkeley

Berkeley, CA

Bachelor of Arts in Applied Mathematics and Data Science

Aug. 2024 – May 2026

- GPA: 3.885/4.000, Cum Laude
- **Relevant Coursework:** Bayesian Inference & Hypothesis Testing (DATA C102), Probability Theory (STAT 134), Data Cleaning & Visualization (DATA C100), Machine Learning (CS 188), Data Privacy & Ethics (DATA C104)

De Anza College

Cupertino, CA

Associate in Science for Transfer in Mathematics and Computer Science

Aug. 2022 – May 2024

- GPA: 4.000/4.000, Summa Cum Laude

PROJECTS & EXPERIENCE

COPD Prevalence Analysis | *Python, Bayesian Inference, PyMC, Altair, Data Filtering*

Dec. 2025

- Cleaned, merged, and analyzed multiple public health datasets, then developed a Bayesian hierarchical model to explain state-level variation in COPD prevalence.
- Estimated effects of smoking, diabetes, poverty, and air pollution on COPD prevalence using MCMC-based posterior inference with PyMC.
- Built Altair visualizations to communicate state-level trends and model results, and ran sensitivity analyses under multiple prior specifications to evaluate model robustness.
- Benchmarked the hierarchical model against an OLS baseline, improving prediction accuracy by reducing RMSE from 0.78 to 0.56 (~28% improvement).

Higgs Boson Discovery Replication Study | *Python, Hypothesis Testing, Data Filtering*

Dec. 2025

- Replicated a high-energy physics analysis using CERN Open Data with event selection criteria based on the original ATLAS research papers.
- Performed hypothesis testing using the profile likelihood ratio and computed statistical significance ($p \approx 2.05 \times 10^{-3}$) for signal detection.
- Documented and justified each data filtering and event-selection step to ensure a reproducible analysis pipeline, addressing gaps in the original publication's methodology.

LLM Multi-Agent System | *JavaScript, Node.js, Gemini API, Google Cloud, PM2, Data Pipeline*

May. 2026

- Built and deployed a Discord-based multi-agent system of 20 AI agents to study emergent conversational behaviors, generating ~7,500 messages of data daily on a Google Cloud Linux VM.
- Engineered a real-time ETL pipeline that collected, cleaned, and structured message data and metadata (author, replies, mentions, reactions, attachments) into an analysis-ready dataset for studying behavioral patterns.
- Deployed and monitored the system on Google Cloud using PM2, ensuring 24/7 uptime with automatic recovery from runtime failures.

Secure File Sharing System | *Go, Cryptography, PKI, HMAC, AES*

Apr. 2026

- Designed and implemented a cryptographically secure file storage and sharing system in Go, supporting user authentication, file operations, and access control over an untrusted datastore adversary.
- Applied public-key encryption (PKE), digital signatures, HMAC, and symmetric-key encryption (AES) to guarantee confidentiality and integrity of all user data, even against a malicious server with full read/write access.
- Engineered a tree-based access revocation mechanism ensuring revoked users immediately lose access even in an untrusted server environment.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, C++, Go, MATLAB, $\text{\LaTeX}$

Data Analysis & Visualization: pandas, NumPy, SciPy, Tableau, Altair, Matplotlib, Seaborn

Statistical Modeling & ML: PyMC (Bayesian Inference), PyTorch, TensorFlow, OpenCV

Tools & Platforms: Excel, PowerPoint, Jupyter Notebook, Google Cloud, Git, VS Code, IntelliJ IDEA

Technologies & APIs: Node.js, Discord.js, Gemini API